

VERTEBRAE EAR

The vertebrate ear has two functional units:

1. **Auditory apparatus:** It is concerned with hearing.
2. **Vestibular apparatus:** It is concerned with posture and equilibrium.

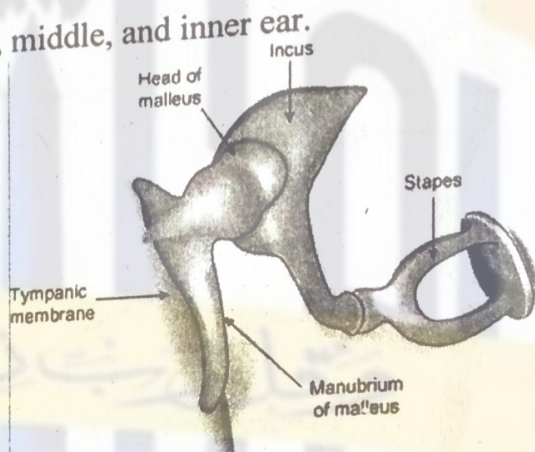
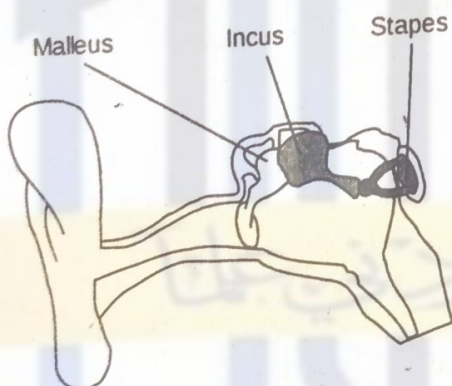
EAR IN MAMMALS

Auditory senses were very important to the early mammals. There are following adaptations in the ear of mammals:

- Auricle:** The ear flap is called auricle.
- Auditory canal or external auditory canal:** They have an auditory tube. It leads into the tympanum. It directs sounds to the middle ear.
- Cochlea:** Cochlea is a long, coiled, sensory structure of the inner ear. It contains receptors for sound. This structure provides more surface area for the receptor cells. Thus mammals have greater sensitivity to pitch and volume other animals.

STRUCTURE OF HUMAN EAR

The human ear has three divisions: the outer, middle, and inner ear.



1. Outer ear

The outer ear consists of the auricle and external auditory canal.

2. Middle ear

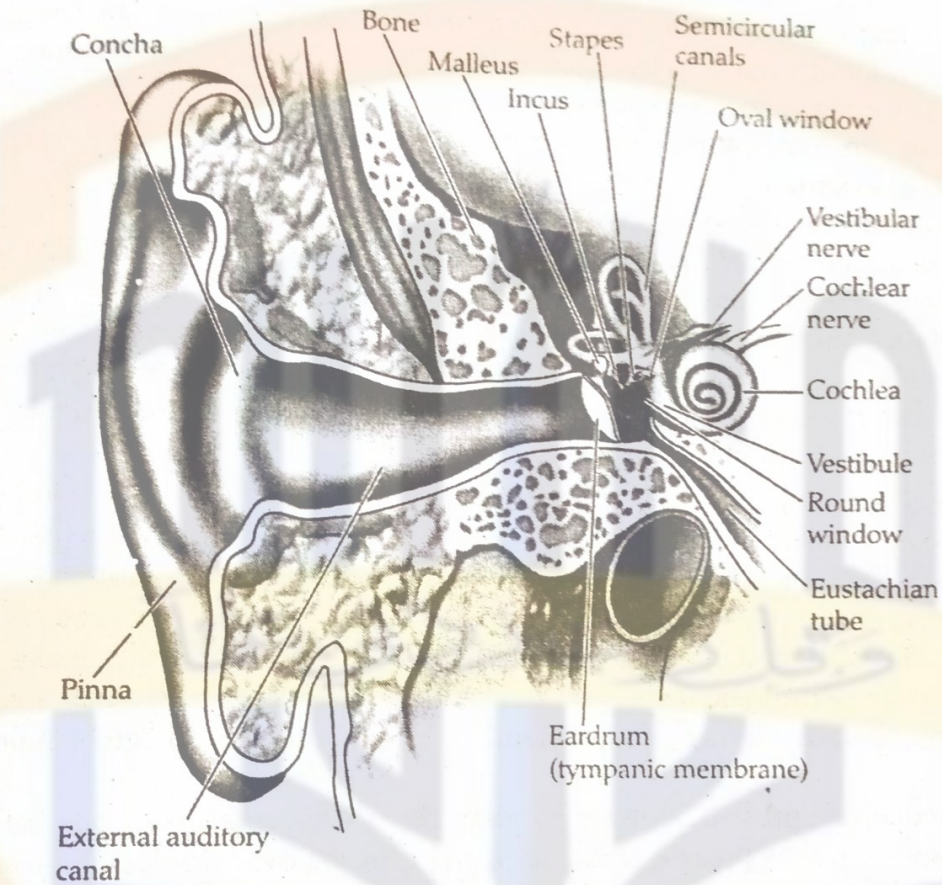
The middle ear begins at the tympanic membrane and ends inside the skull. Two small membranous openings are located in the middle ear. These are called **oval** and **round windows**. Three small ossicles (bones) are present between the tympanic membrane and the oval window. They are named according to their shapes. These ossicles are:

- Malleus (hammer):** It adheres to the tympanic membrane and connects to the incus.
- Incus (anvil):** The incus connects to the stapes.
- Stapes (stirrup):** It adheres to the oval window.

An auditory tube or **Eustachian tube** is present in the middle ear. It extends from the middle ear to the nasopharynx. It equalizes air pressure between the middle ear and the throat.

The major function of the middle-ear ossicles is to transfer sound energy from air to the liquid of the inner ear. Airborne sound striking a liquid surface is almost all reflected. Only about 1/30th of sound energy is transferred to the liquid. The energy transfer is poor because liquids cannot be compressed. Hence they have a low volume of movement in response to sound pressure. These considerations mean that for significant transfer of energy from the eardrum to the liquid medium of the inner ear, the pressure developed at the inner ear must be increased. The middle-ear ossicles achieve this increase in pressure

by applying forces from a relatively large area (55 mm² for human eardrum) onto a much smaller area (3 mm² for foot plate of the stapes). The concentration of force on a small area provides the necessary increase in pressure. It allows efficient transfer of sound energy.



b) The inner ear

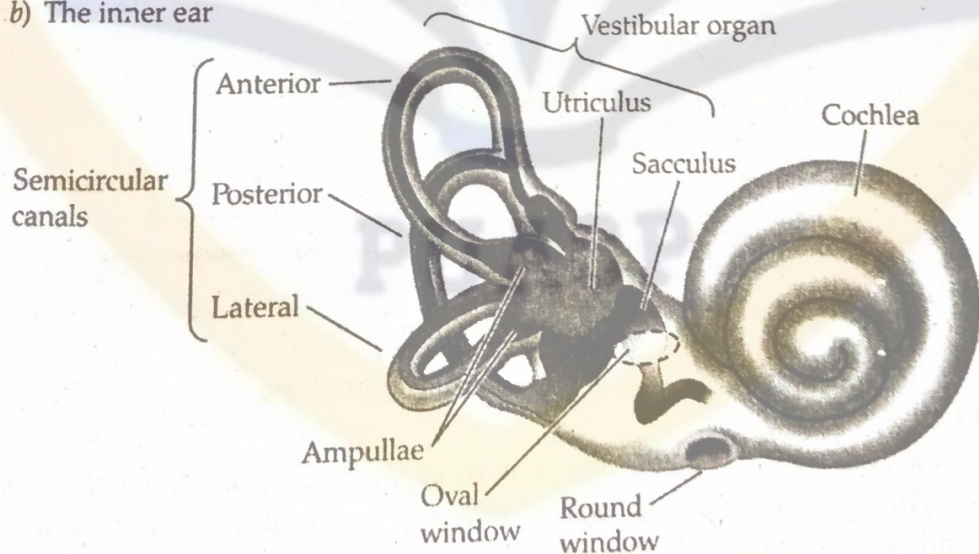


Fig: Structure of Human Ear

3. Inner ear

The inner ear has two parts: Vestibular organ and cochlea.

- a) **Vestibular organ:** It is composed of vestibule and three semicircular canals. This organ is concerned with equilibrium. Each vestibular organ comprises three semicircular canals. These canals detect angular acceleration of the head and body. It has two otolith organs. These are called the **sacculus** and **utricle**. They detect linear movement and acceleration.
- b) **Cochlea:** The third component is cochlea. It is involved in hearing.

Structure of cochlea

Basilar Membrane

The cochlea is a coiled tube containing chambers. These chambers are filled with fluid. A membranous structure **basilar membrane** separates cochlea into an upper chamber called **scala vestibuli** and a lower chamber called **scala tympani**. The oval window moves. It creates fluid movement in the cochlea. This movement vibrates the basilar membrane and stimulates auditory hair cells that sit on it. The basilar membrane varies in width and thickness along its length:

- It is narrow, thick, and rigid at the base of the cochlea near the oval window.
 - It is wider, thinner, and more flexible at the apex, farthest from the oval window.
- Sound waves push in at the oval window. They set up traveling pressure waves in the fluid-filled cochlea. The pressure waves produce minute movements of basilar membrane. Basilar membrane has differences in its width and rigidity along its length. It gives the basilar membrane a variable mechanical advantage. As a result sound waves of different frequencies vibrate basilar membrane maximally at different points along its length.

- a) High-frequency pressure waves vibrate the stiffer, narrower part of the basilar membrane. It part is present at the basal end near the oval window.
- b) Low-frequency sounds vibrate basilar membrane at its broader apical end.

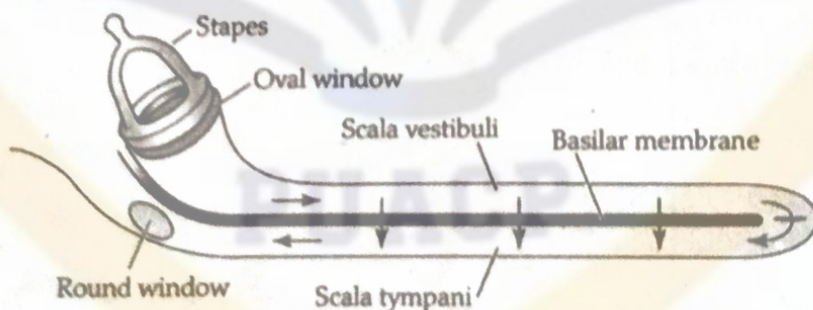


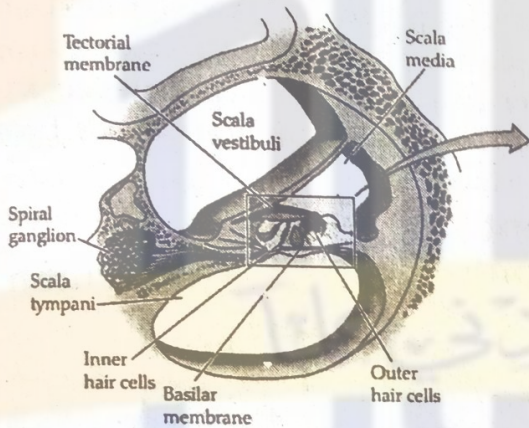
Fig: Anatomy of the cochlea

Organ of Corti

The hair cells and various accessory structures of the organ of Corti sit on the basilar membrane. They vibrate up and down with it. Stereocilia of hair cells project into a separate fluid compartment of cochlea called **scala media**. The hair cells of the organ of Corti are of two kinds. There are three rows of outer hair cells and a single row of inner

hair cells. The hair cells are covered by a flap of tissue called **tectorial membrane**. The stereocilia of hair cells are very close to or in contact with the tectorial membrane. The basilar membrane moves up and down. It causes the stereocilia to push up against the tectorial membrane. Hence the hair bundle is displaced. Displacement in one direction depolarizes hair cell membrane potential. The displacement in the opposite direction hyperpolarizes it. Both inner and outer hair cells depolarize in response to tones. The amount of depolarization depends on sound intensity and frequency. Auditory hair cells make synaptic contact with afferent neurons of the **auditory (cochlear) nerve** and also receive **efferent synapses**. Most (80–95%) of afferent neurons form synapse with inner hair cells. Thus the inner hair cells are the major source of auditory input to the brain.

(a) A cross section through the cochlea



(b) The organ of Corti

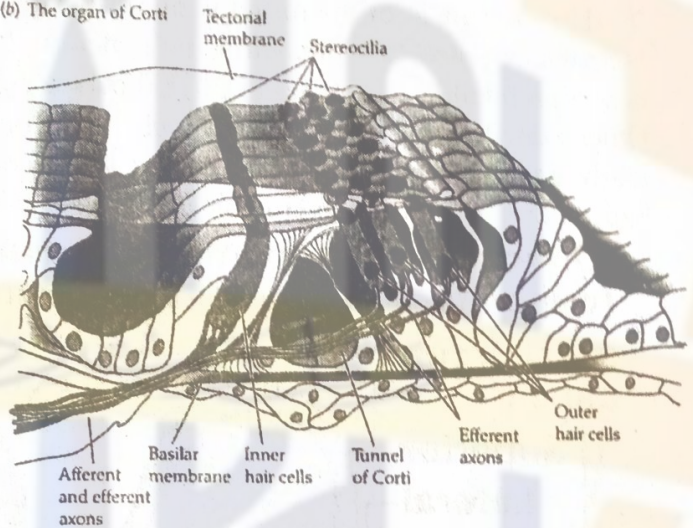
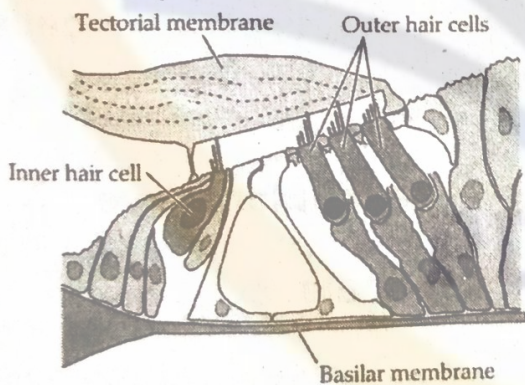


Fig: Cross section of Cochlea and Organ of Corti

(a) Hair cell displacement



Displacement of the basilar membrane bends the stereocilia of inner and outer hair cells.

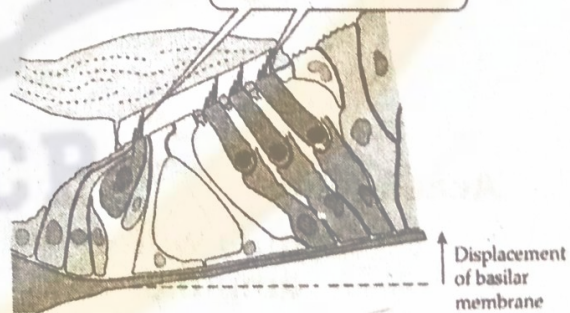


Fig: Hair cells Displacement

Process of hearing

Following steps are involved in the process of hearing.

1. Sound waves enter the outer ear. It creates pressure waves. These pressure waves reach the tympanic membrane.
2. The pressure waves vibrate the tympanic membrane. The vibrations move the

Malleus on the other side of the membrane.

3. The handle of the malleus articulates with the incus and vibrate it.
 4. The vibrating incus moves the stapes back and forth against the oval window.
 5. The movements of the oval window set up pressure changes. It vibrates the fluid in the inner ear. These vibrations are transmitted to the basilar membrane. The basilar membrane starts vibrating.
 6. **Organ of Corti** is present on the basilar membrane. Organ of Corti is hair cells. They contact with the overlying **tectorial membrane**. This membrane bent. It produces a generator potential. It causes an action potential. This action potential travels along the **vestibulocochlear nerve** and reach the brain for interpretation.
 7. The movement of the round window dissipates the vibrations in the cochlear fluid.
- Humans can hear low-pitched sounds, below 20 cycles per second. Young children can hear high-pitched sounds up to 20,000 cycles per second. This ability decreases with age. Other vertebrates can hear sounds at much higher frequencies. For example, dogs can easily detect sounds of 40,000 cycles per second. Thus, dogs can hear sounds from a high-pitched dog whistle. Human cannot hear this whistle.

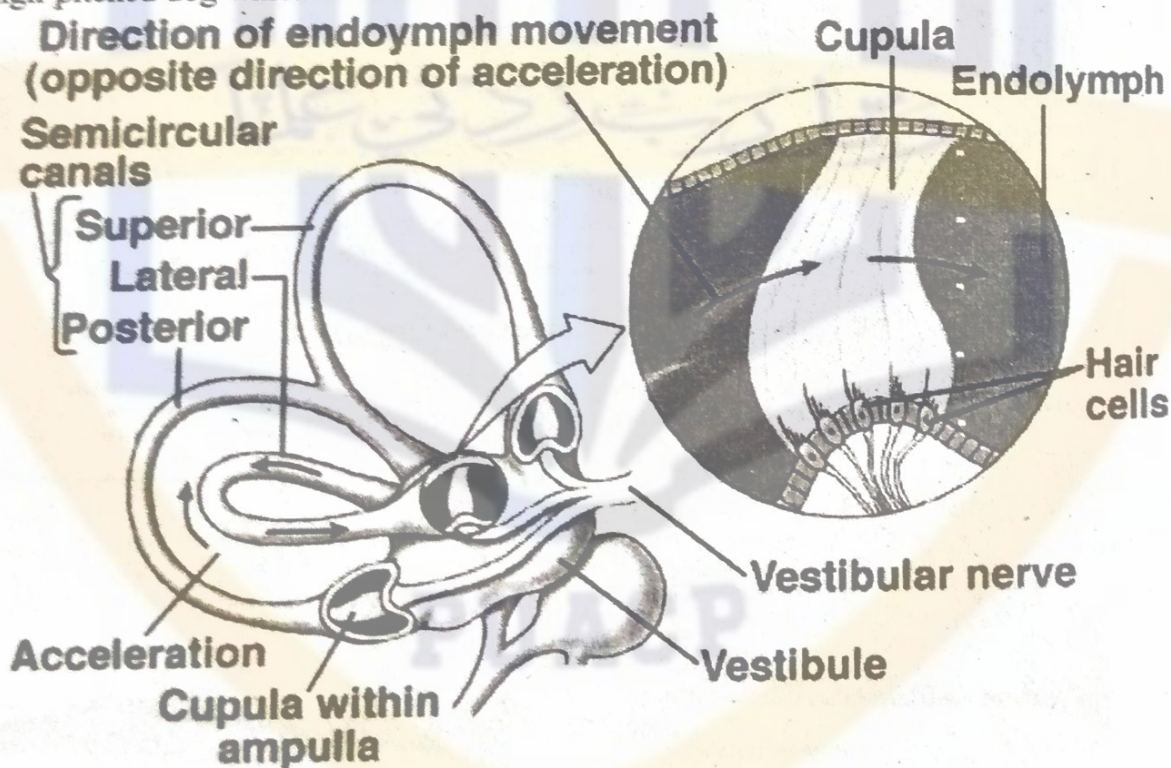


Fig: Static Equilibrium (balance)

Mechanism of equilibrium

The sense of equilibrium is divided into two separate senses.

- (a) **Static equilibrium.** The sensing of movement in one plane (either vertical or horizontal) is called static equilibrium.
- (b) **Dynamic equilibrium:** The sensing of angular and rotational movements is called

dynamic equilibrium.

The semicircular canals have otoliths (cupula). The **otolith** has sensory granules. The semicircular canals rest on hair cells when the body is still. The granules are displaced when the head or body moves horizontally, or vertically. It displaces the gelatinous material. This displacement bends the hairs slightly. Thus the hair cells initiate a generator potential and action potential. Continuous movement of the fluid in the semicircular canals causes motion sickness or seasickness in humans.

FACTORY (CHEMORRECEPTORS) AND ...

PHOTORECEPTORS (VISION)

The receptors which are sensitive to light are called photoreceptors. All photoreceptors possess light-sensitive pigments. These pigments are **carotenoids** and **rhodopsin**. These pigments absorb photons of light energy and produce a generator potential. Different types of photoreceptors are present in different animals.

Type of Photoreceptor Cells

Photoreception arose early in evolution. All organisms detect light by using a pigment. These pigments are called **photopigment**. All animals use one dominant photopigment, **rhodopsin** for photoreception. The photopigment is associated with cell membranes.

Hence all photoreceptor cells have greatly increased membrane surface areas that increase their light sensitivity. Photoreceptor cells are subdivided into ciliary and rhabdomeric.

a) **Ciliary photoreceptor cells:** In these cells, modified cilia contain rhodopsin molecules. Vertebrate photoreceptor cells are ciliary.

b) **Rhabdomeric photoreceptor cells:** These cells have collections of microvilli that increase the membrane surface area. Arthropod photoreceptor cells are rhabdomeric. Many other phyla have examples of both photoreceptor types.

Types of Eyes

Most animal phyla contain light-responsive sense organs called eyes. These eyes differ in type and complexity. Eyes are sense organs that contain photoreceptors mediating light responses. Eyes may also form images of visual world. There are two major kinds of image forming eyes.

a) **Camera eye:** In this eye, a lens forms an inverted image on photoreceptors at the back of the eye.

b) **Compound eye:** This eye has many facets called **ommatidia** (singular **ommatidium**). Each ommatidium has its own lens. The ommatidia collectively form image called a **mosaic image**. Each ommatidium conveys information about one part of the visual world. The nervous system forms an image a mosaic image.

VERTEBRATE EYE

All vertebrates have camera eyes. In this type of eye, the cornea and lens focus an inverted image of visual field on the **retina**.

The eyeball has three layers: sclera, choroids and retina.

1. **Sclera:** It is the outermost layer of eye. The transparent **cornea** covers the front of the eyeball. It is continuous with the sclera
2. **Choroids:** It is a thin middle layer. Choroid tissue also extends to the front of the eyeball. They form the **iris**, **ciliary body**, and **suspensory ligaments**. The iris is colored. It has light-screening pigments. It also has radial and circular smooth muscles. These muscles regulate the amount of light entering the pupil.
3. **Retina:** It is the innermost layer. It contains many photoreceptors.

There are two chambers of eyes:

- (a) **Aqueous humor:** It is a clear fluid. It fills the anterior chamber of eye. The chambers lie between the lens and the cornea. The **lens** is present behind the iris,
- (b) **Vitreous chamber:** It is a jelly like vitreous body. It fills chamber behind the lens.

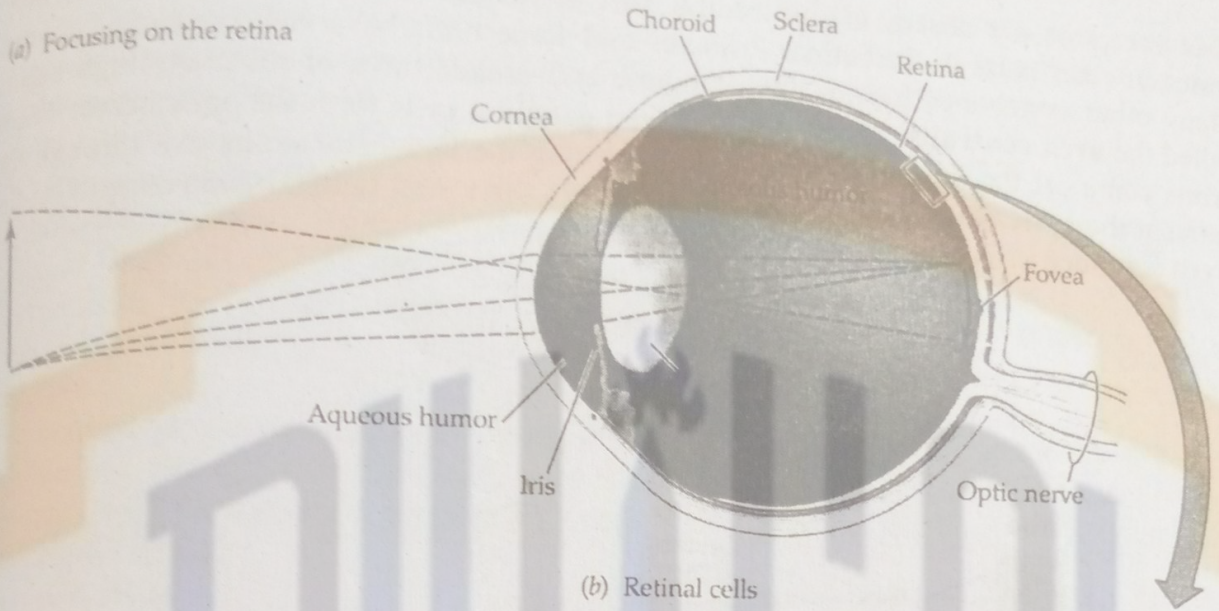
Retina

Retina is photoreceptor containing layer at the back of the eye. The retina of the vertebrate eye is a developmental outgrowth of the brain. It contains **rod** and **cone** photoreceptor cells. It also has network of neurons, horizontal cells, bipolar cells, amacrine cells, and ganglion cells. A pigmented epithelium lies at the back of the retina. It absorbs light not captured by the photoreceptors. It controls ionic environment around the rods and cones and synthesis of 11-cis retinal. The retina is inverted. Its photoreceptors are in the outermost layer.

Fovea

The light must pass through all retinal layers. Then it reaches the outer segments of the rods and cones. The retinal layers do not degrade the image. Many retinas have a central high-acuity region. In this region, intervening cell layers and blood vessels are displaced to side. In humans this region is **fovea**. It is a depression 1.5 millimeters (mm) in diameter (5° of visual angle). The central 1° of fovea contains tightly packed cones. Rod

(a) Focusing on the retina



(b) Retinal cells

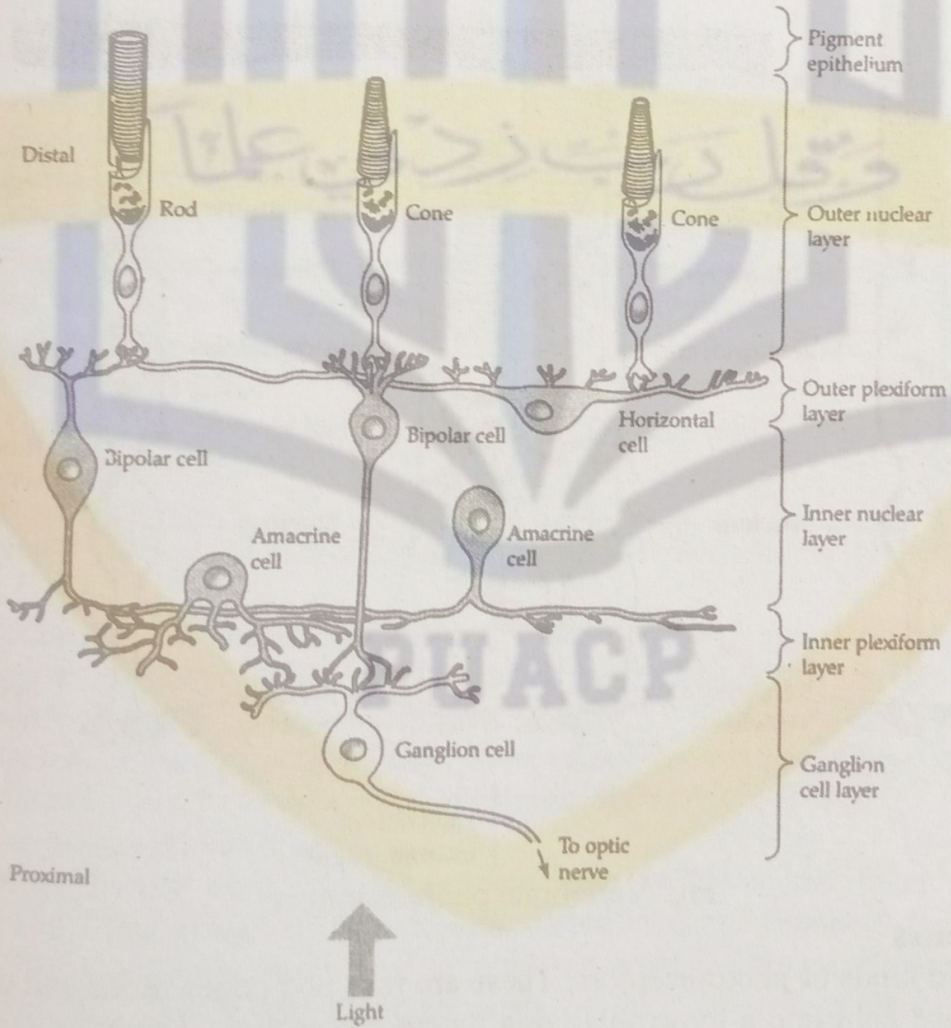


Fig: Structure of the mammalian eye and retina

photoreceptors are absent in this central part of human fovea. But greatly outnumber cones in other parts in the retina. Primates and some birds have well-developed foveas. Many other vertebrates have a less elaborate and broader area of relatively high acuity called the **area centralis**. The axons of retinal ganglion cells form the optic nerve. These axons come off the inner side of the retina, facing the lens. The axons exit through the retina at the **optic disc**. It produces a **blind spot** in the visual field. Human depend on the fovea for much of our high-acuity vision.

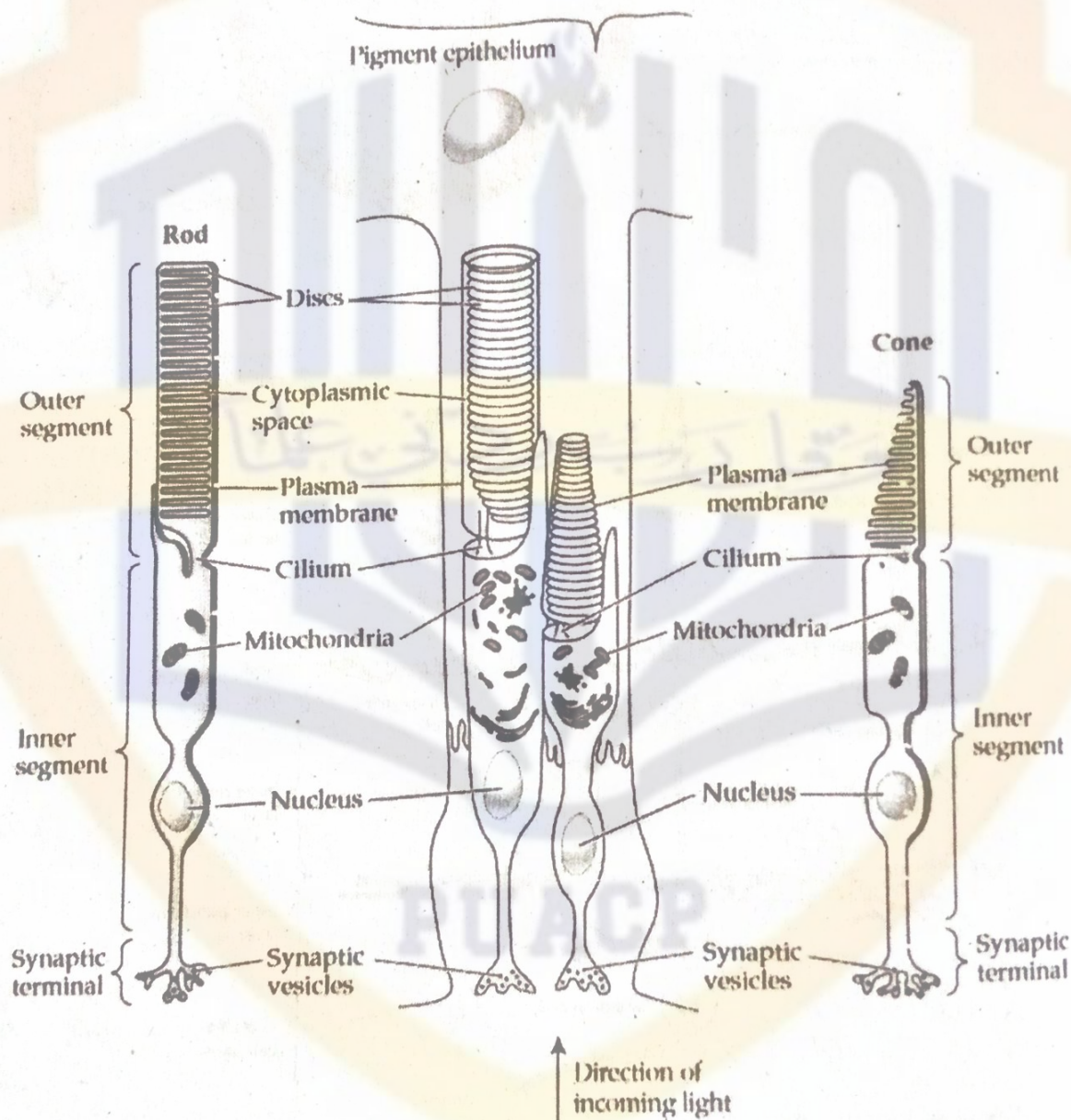


Fig: Vertebrate photoreceptors

Rod and Cones

Retina has two kinds of photoreceptors. These are rods and cones. Rods and cones of the retina transduce light into a hyperpolarizing receptor potential.. The more-sensitive rods

are used in dim light. The cones are used in brighter light for color vision, and for high-acuity vision in humans. **Nocturnal** animals tend to have retinas in which most or all photoreceptors are rods. But cones predominate in retinas of strongly diurnal animals.

Structure of rods and cones

Rods and cones are composed of two segments

- a) **Outer segment:** Both rods and cones have an outer segment. It contains photosensitive membranes. The outer segment is derived from a modified cilium.
- b) **Inner segment:** Their inner segment contains nucleus, mitochondria, other cell organelles, and synaptic terminal. The inner and outer segments are connected by a short ciliary stalk.

Lamellae

The photoreceptor outer segments contain many flattened lamellae of membranes. These lamellae are derived from the cell membrane. In the cones, these lamellae retain continuity with the outer cell membrane. Hence lumen of each lamella is continuous with extracellular space.

Flattened Discs

In rods, lamellae become separated from the outer membrane. They form internalized **flattened discs**. Several hundred to 1000 discs stacked like pancakes. They fill the rod outer segment. The membranes of rod discs and cone lamellae contain photopigment **rhodopsin**. The disc membranes and cell membrane are discontinuous in rods. Hence light-induced changes in rhodopsin at the disc membrane must affect the ion permeability of outer membrane. An electrical change in the disc membrane will not directly affect surface membrane. Instead, there is an **intracellular messenger**. This messenger conveys a change from discs to outer surface membrane. This intracellular messenger is cyclic **GMP (cyclic guanosine monophosphate, or cGMP)**.

Mechanism of Transduction

The transduction of light into an electrical signal in rod and cone photoreceptors has four stages:

1. Light activates rhodopsin.
2. The activated rhodopsin stimulates a G protein. It activates a phosphodiesterase enzyme.
3. Enzyme decreases the concentration of cyclic GMP in the photoreceptor cytoplasm.
4. The decrease in cyclic GMP closes cyclic nucleotide-gated ion channels. The photoreceptor channels close. Therefore, Na^+ influx decreases and photoreceptor **hyperpolarizes** rather than depolarizes. Most of our knowledge of phototransduction comes from rods. But the mechanisms are similar for both.

Mammalian transduction cascade

1. **In darkness:** In the dark, **cGMP** concentration within a rod outer segment is relatively high. The cGMP binds to cyclic nucleotide-gated channels. It opens these channels. Hence the permeability of the outer segment membrane becomes high for cations like Na^+ . The cGMP-gated channels of vertebrate photoreceptors are not voltage-gated. Instead, they open when they bind cGMP on cytoplasmic surface.

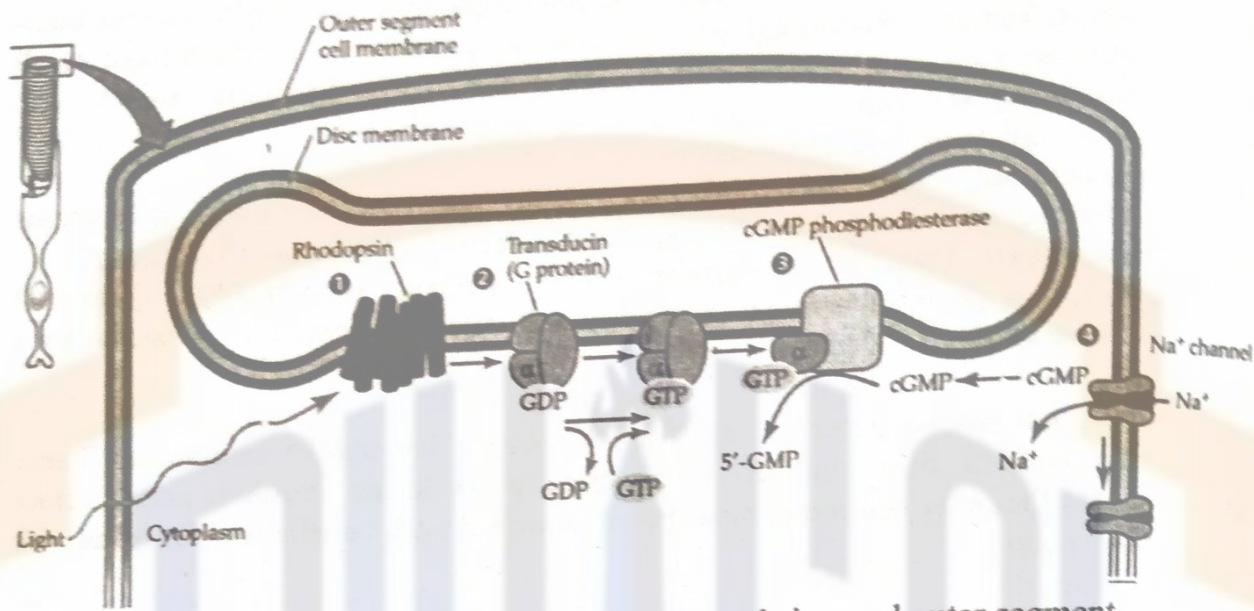


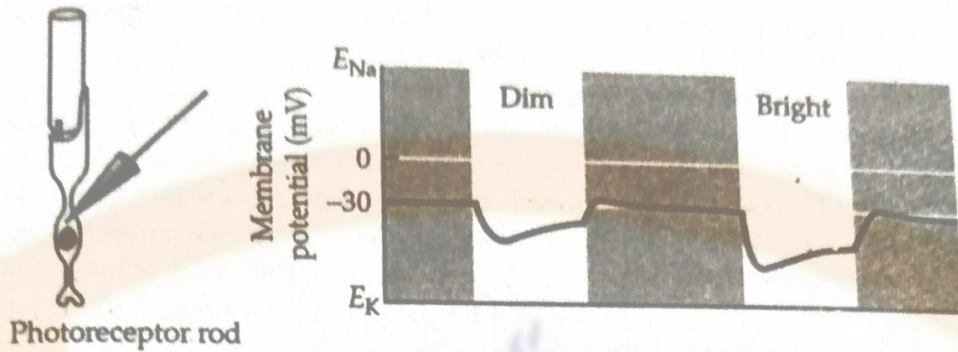
Fig: Phototransduction closes cation channels in a rod outer segment

2. **In light:** The light-induced change in rhodopsin activates a series of reactions at the disc membrane. It results in an enzymatic degradation of cGMP.

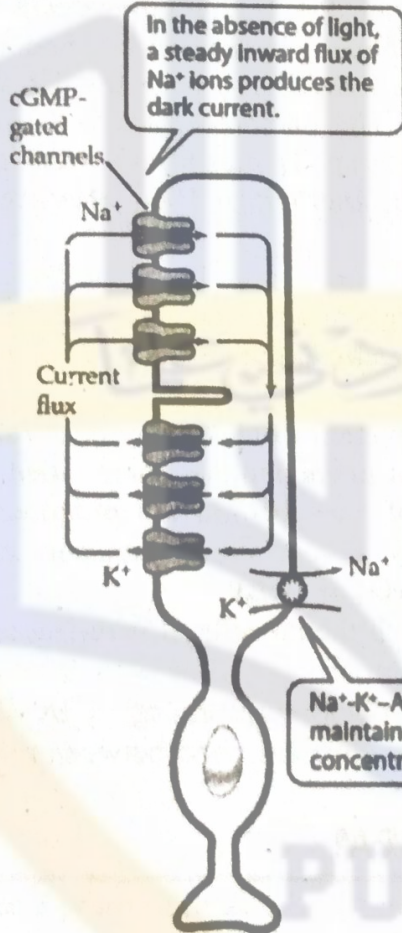
- Light-stimulated rhodopsin (metarhodopsin II) activates **G protein transducin**.
- The activated G protein stimulates **cGMP phosphodiesterase (PDE)**. PDE is an enzyme in the disc membrane. It hydrolyzes cGMP to 5'-GMP.
- The activated PDE decreases cytoplasmic concentration of cGMP. Thus the cation channels close. Thus second messenger-mediated response (cGMP) to light in a rod outer segment is decreased. It decreases Na⁺ influx. It produces **hyperpolarization**. According to Goldman equation the membrane potential depends on the ratio of permeabilities to Na⁺ and K⁺. The resting Na⁺ permeability of a rod or cone is higher than in a normal neuron. Therefore, membrane potential is less negative. In fact, resting membrane potential of a rod or cone is about -30 millivolts (mV). It is much more depolarized than the resting membrane potential of a typical neuron.

3. **Dark current:** This constant flow of Na⁺ into the cell produces a **dark current**. It is constantly counteracted by expenditure of ATP to pump Na⁺ out of the cell with a very active Na⁺/K⁺ ATPase pump. This pump is located in the inner segment. The dark current keeps the rod relatively depolarized.

4. **Hyperpolarization:** The light induced decrease in cytoplasmic concentration of cGMP shifts the equilibrium of cGMP binding to channel protein. Hence cGMP dissociates from channel and the channel closes. The response to light is thus a decrease in the dark current. It causes the membrane potential to become more negative. Thus it produces a **hyperpolarizing receptor potential**. This potential is graded according to the intensity of the light flash.



(b) Dark



(c) Light

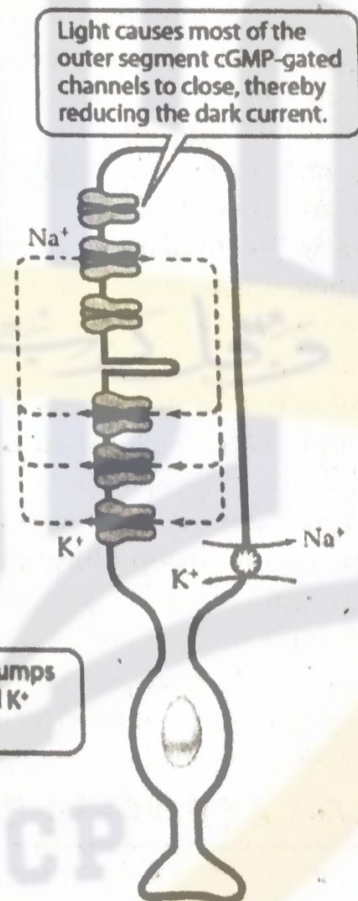


Fig: Light hyperpolarizes vertebrate photoreceptors.

Vertebrate rods can produce a detectable change in dark current from absorption of a single photon. Each photon absorbed by rhodopsin leads to approximately 10^6 Na^+ ions not entering a rod outer segment. The activation of one rhodopsin molecule can lead to the activation of many transducin and phosphodiesterase molecules. It causes the enzymatic degradation of a large number of cGMP molecules. This decreased cGMP

concentration can close many Na^+ channels. It blocks the entry of enough Na^+ ions to produce a change in membrane potential of order of 1 mV.

Enzymatic regeneration of rhodopsin

If rods are stimulated by very bright light, the photoreceptors recover very slowly. This is because bright light has converted 11-cis retinal to all-trans retinal in a large fraction of pigment of eye. This process called **bleaching**. After a bleach greater than about 1% of photopigment, recovery of rod vision is prolonged. This slow adjustment to darkness is called **dark adaptation**. Complete recovery of sensitivity after bleaching illumination requires the regeneration of the all-trans retinal back to 11-cis retinal. Regeneration can occur either photochemically or enzymatically.

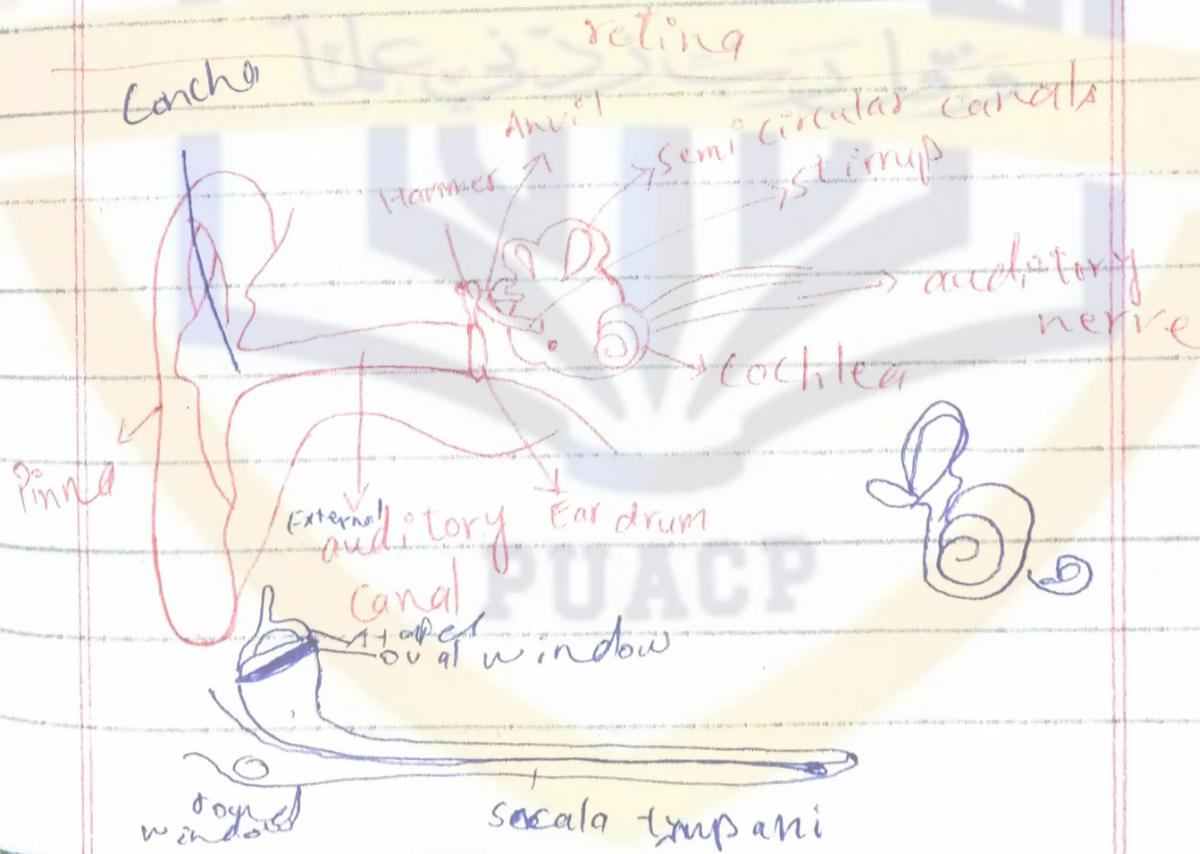
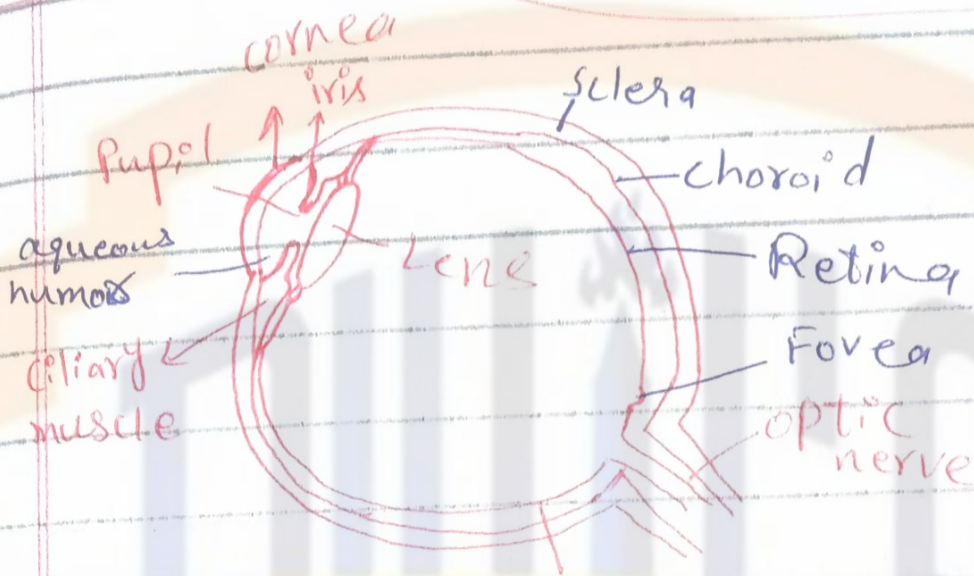
(a) Photochemical Mechanism

In insects, most regeneration occurs photochemically. The metarhodopsin with all-trans retinal can absorb a second photon. It converts directly back to 11-cis rhodopsin. This has obvious advantages but one disadvantage. The light illuminating the eye will generally stimulate both rhodopsin and metarhodopsin. The photoreceptors will usually contain some combination of both pigments. Many arthropods start afresh every day, breaking down and resynthesizing the light-sensitive microvilli. Then they regenerate the rhodopsin on a circadian basis.

(b) Enzymatic Mechanism

For vertebrate rods, regeneration occurs by an enzymatic mechanism.

1. After bleaching of 11-cis retinal to all-trans retinal, the all-trans retinal unbound from opsin protein.
2. The all-trans retinal (aldehyde) is then converted to all-trans retinol (alcohol) by an enzyme **retinol dehydrogenase**. It is present in the photoreceptor called.
3. The retinol is then transported to layer of cells behind the photoreceptors called **pigment epithelium**. It is transported by a specialized transport protein called **interphotoreceptor retinoid binding protein, or IRBP**.
4. The all-trans retinol is taken up by pigment epithelial cells. It is re-isomerized back to 11-cis retinol and then converted to 11-cis retinal.
5. The 11-cis retinal is transported back to photoreceptors again by IRBP. It then recombines with opsin by reformation of the covalent bond between retinal aldehyde and amino group of lysine in opsin.



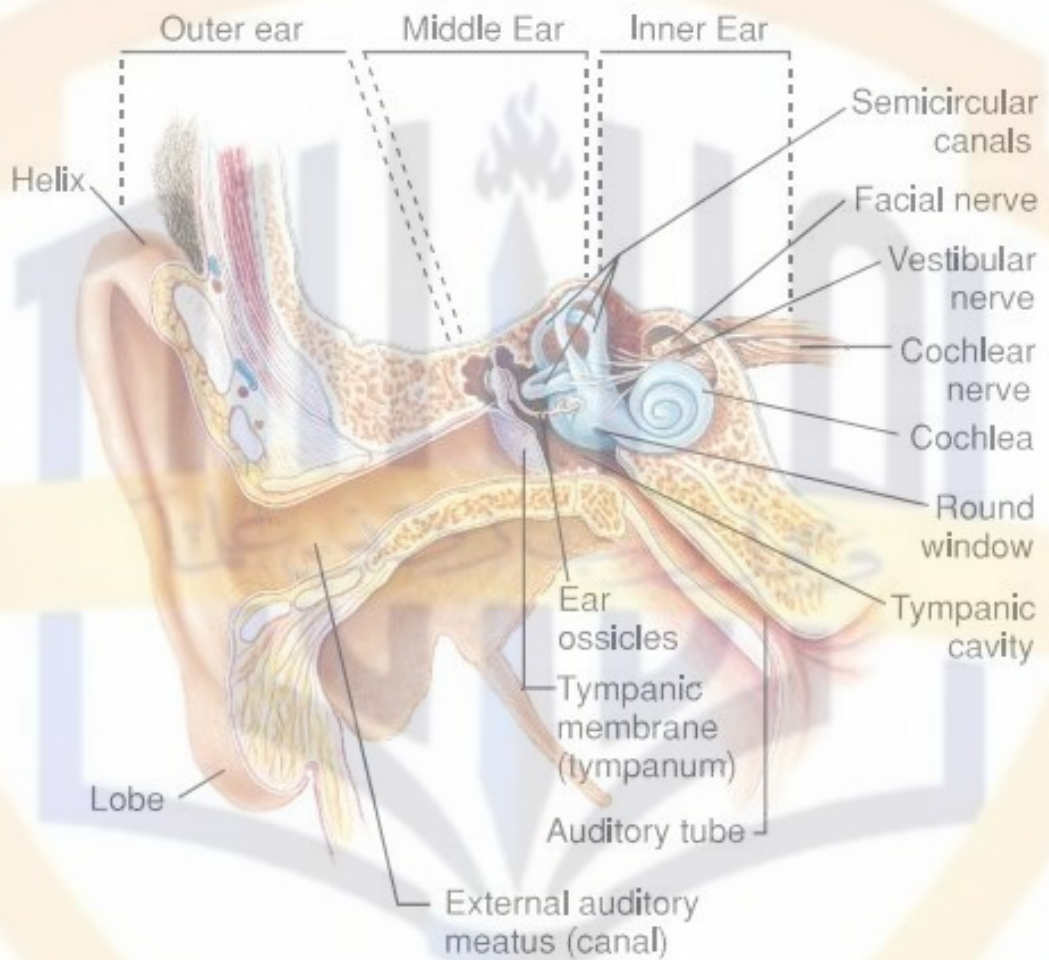


FIGURE 24.21

Anatomy of the Human Ear. Note the outer, middle, and inner regions. The inner ear includes the semicircular canals, which are involved with equilibrium, and the cochlea, which is involved with hearing.

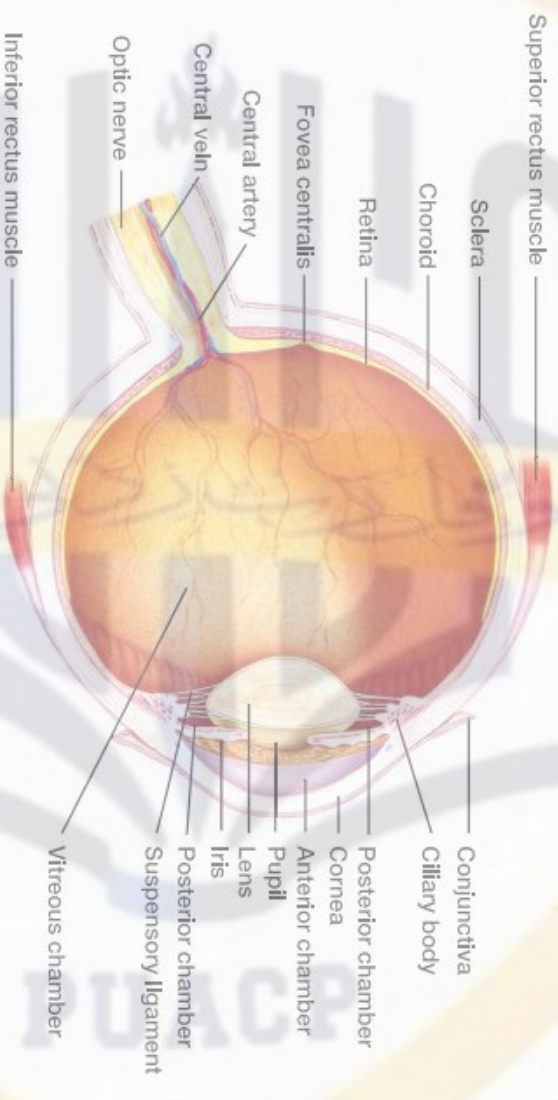


FIGURE 24.29

Internal Anatomy of the Human Eyeball. Light passes through the transparent cornea. The lens focuses the light on the rear surface of the eye, the retina, at the fovea centralis. The retina is rich in rods and cones.